

JEE MAIN REVISION MASTER: SOLID STATE 2026

0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 23 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Solid State - Core Application	23 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Solid State - Core Application (Very High Yield — 23 PYQs)

Core Principle: Key problem pattern in Kcet Solid State.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which of the following crystals has the unit cell such that $a=b \neq c$ and $\alpha=\beta=90^\circ$, $\gamma=120^\circ$?..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "The number of atoms in 4.5 g of a face-centred cubic crystal with edge length 300 pm is (Given : Density = 10 g cm^{-3} and $N_A = 6.022 \times 10^{23}$)." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "A metal crystallises in a body centred cubic lattice with the metallic radius $\sqrt{3}r$. The volume of the unit cell in Å^3 ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

The number of atoms in 4.5 g of a face-centred cubic crystal with edge length 300 pm is (Given : Density = 10 g cm^{-3} and $N_A = 6.022 \times 10^{23}$)

- (A) 6.6×10^{20}
- (B) 6.6×10^{23}
- (C) 6.6×10^{19}
- (D) 6.6×10^{22}

Correct Answer: None

Solution Analysis: Given, Face centred crystal, so $Z=4$ edge length $(a)=300 \text{ pm}=300 \times 10^{-10} \text{ cm}$ density $(d)=10 \text{ g/cm}^3$ $N_A=6.022 \times 10^{23}$ Using the formula $d = \frac{Z \times M}{N_A \times a^3}$ $M = \frac{d \times N_A \times a^3}{Z} = \frac{10 \times 6.022 \times 10^{23} \times (300 \times 10^{-10})^3}{4} \Rightarrow M = 40.65 \text{ g}$ So, 40.65 g contains 6.022×10^{23} atoms Let 4.5 g contains x atom $\Rightarrow x = \frac{6.022 \times 10^{23} \times 4.5}{40.65} = 6.6 \times 10^{22}$ atoms